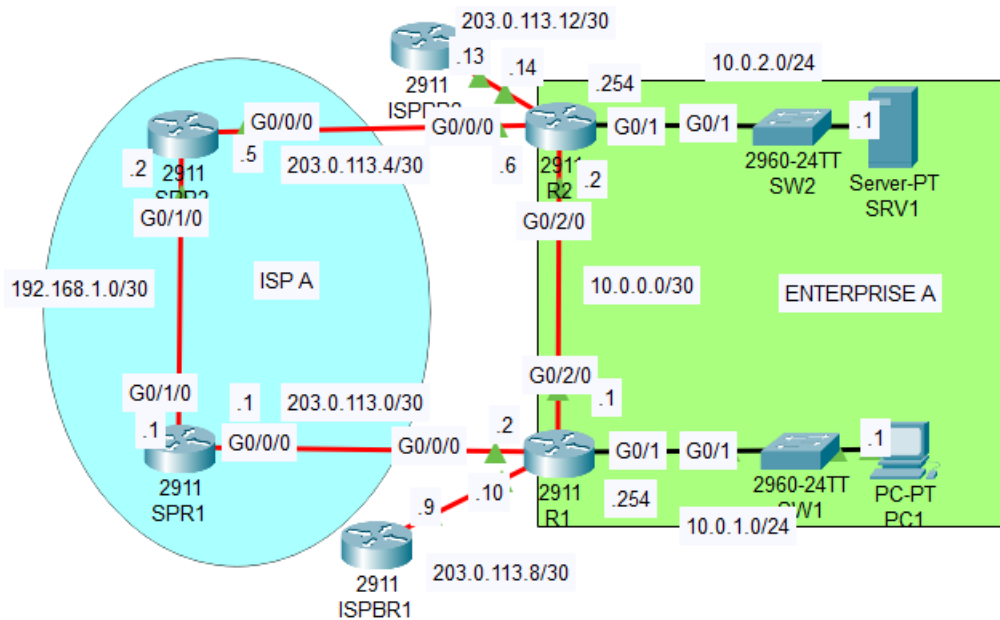




1. Check the routing tables of R1 and R2. Which dynamic routing protocol is Enterprise A using? Which route will be used if PC1 tries to access SRV1? Which route will be used if PC1 tries to access remote server 1.1.1.1 over the Internet? Test by pinging SRV1 and 1.1.1.1

```
R1>en
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 203.0.113.9 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/2/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/2/0
C       10.0.1.0/24 is directly connected, GigabitEthernet0/1
L       10.0.1.254/32 is directly connected, GigabitEthernet0/1
C       10.0.2.0/24 [110/2] via 10.0.0.2, 00:28:36, GigabitEthernet0/2/0
    203.0.113.0/24 is variably subnetted, 4 subnets, 2 masks
C       203.0.113.0/30 is directly connected, GigabitEthernet0/0/0
L       203.0.113.2/32 is directly connected, GigabitEthernet0/0/0
C       203.0.113.8/30 is directly connected, GigabitEthernet0/1/0
L       203.0.113.10/32 is directly connected, GigabitEthernet0/1/0
S*    0.0.0.0/0 [1/0] via 203.0.113.9
```

As shown here, the 10.0.2.0 is marked with O, which means it was learned via OSPF, the routing protocol being used by Enterprise A. If PC1 tries to access SRV1, the most specific matching-route is the OSPF route to 10.0.2.0/24, through G0/2/0. If PC1 tries to access the remote server 1.1.1.1 over the Internet, R1 will use its default gateway route through 203.0.113.9.

```

R2>en
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

```

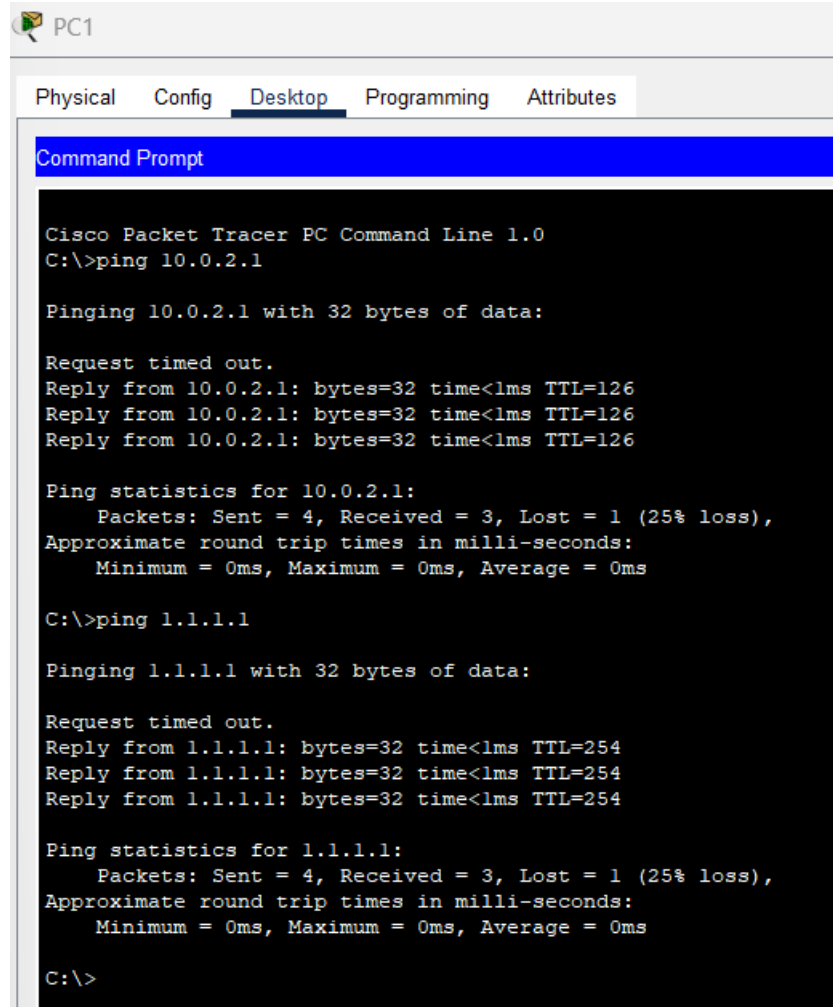
Gateway of last resort is 203.0.113.13 to network 0.0.0.0

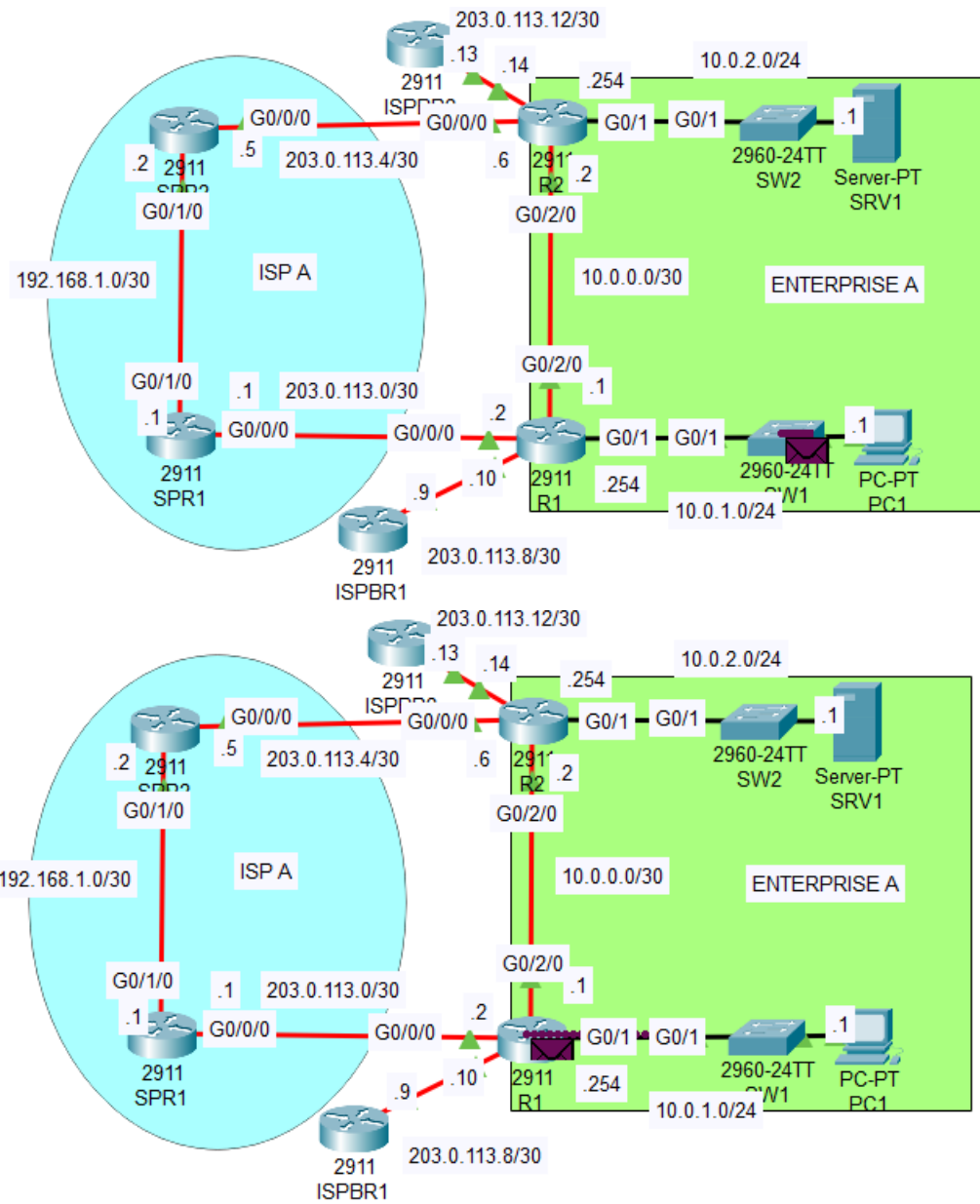
```

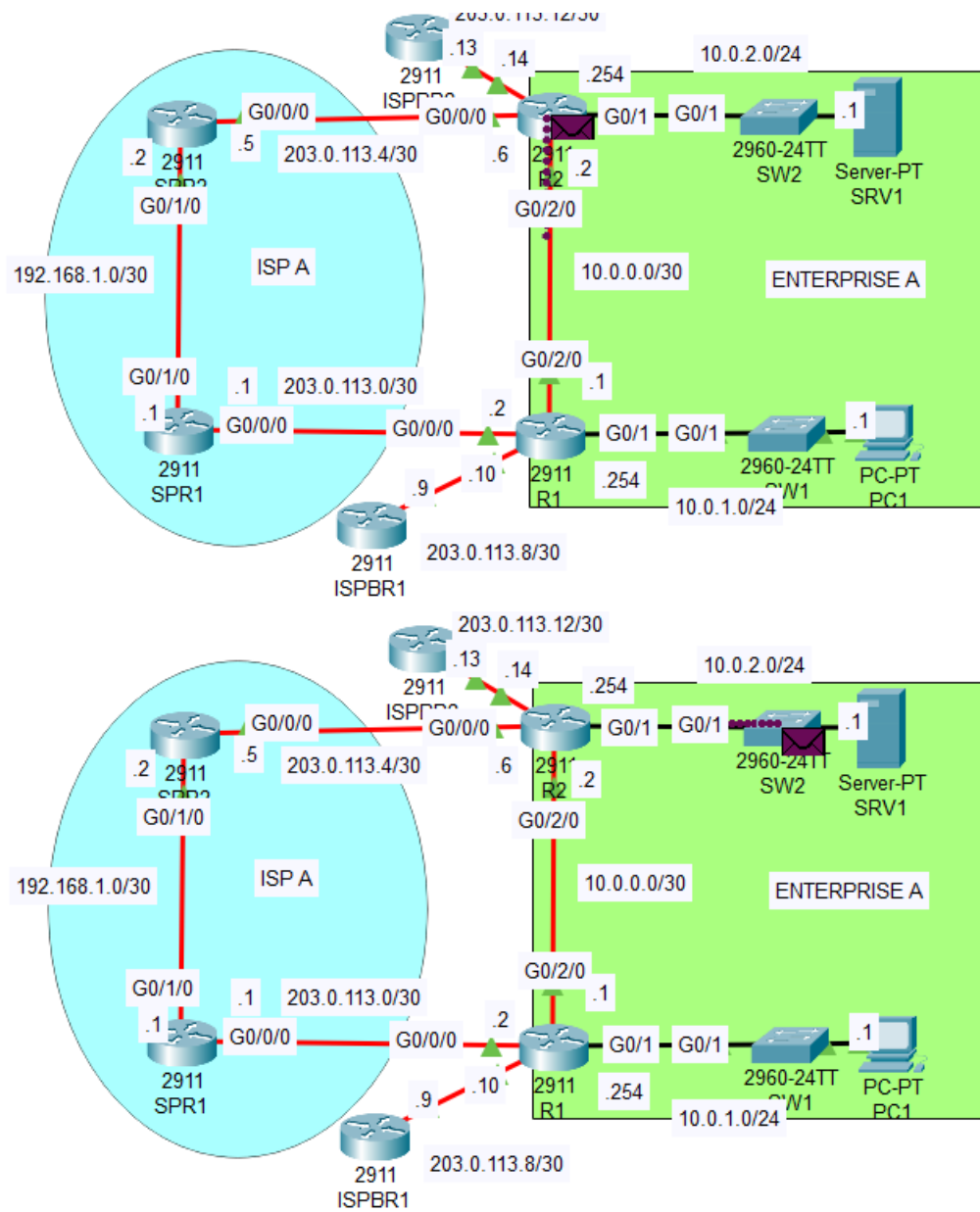
      10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/2/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/2/0
O       10.0.1.0/24 [110/2] via 10.0.0.1, 00:37:07, GigabitEthernet0/2/0
C       10.0.2.0/24 is directly connected, GigabitEthernet0/1
L       10.0.2.254/32 is directly connected, GigabitEthernet0/1
      203.0.113.0/24 is variably subnetted, 4 subnets, 2 masks
C       203.0.113.4/30 is directly connected, GigabitEthernet0/0/0
L       203.0.113.6/32 is directly connected, GigabitEthernet0/0/0
C       203.0.113.12/30 is directly connected, GigabitEthernet0/1/0
L       203.0.113.14/32 is directly connected, GigabitEthernet0/1/0
S*     0.0.0.0/0 [1/0] via 203.0.113.13

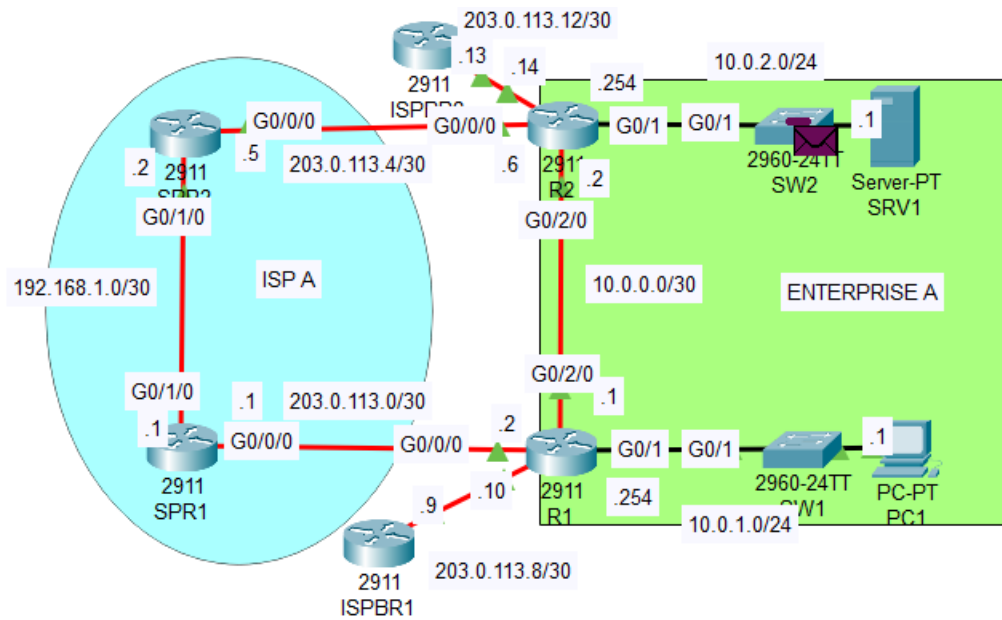
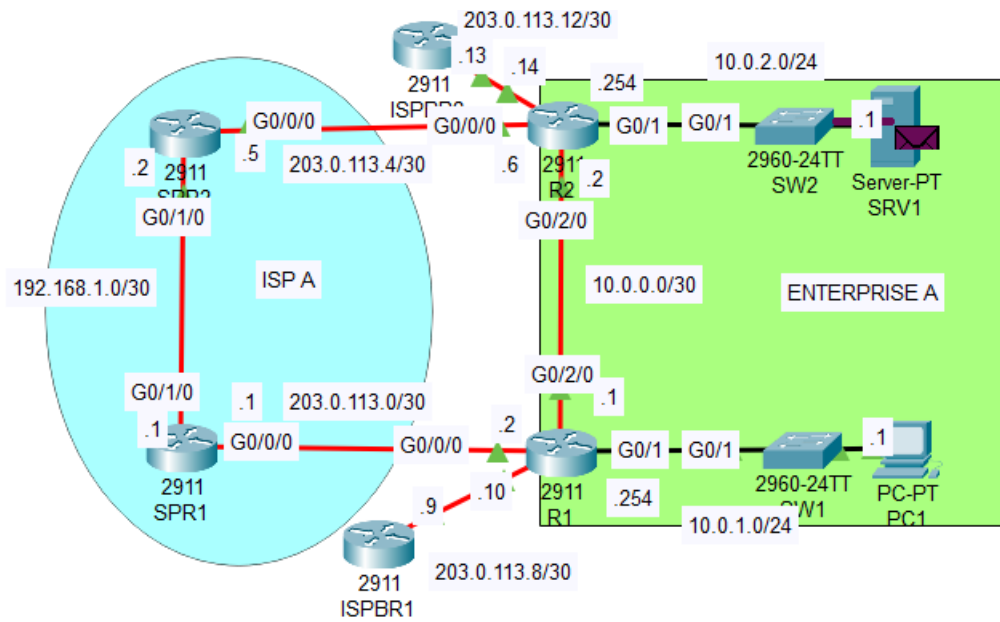
```

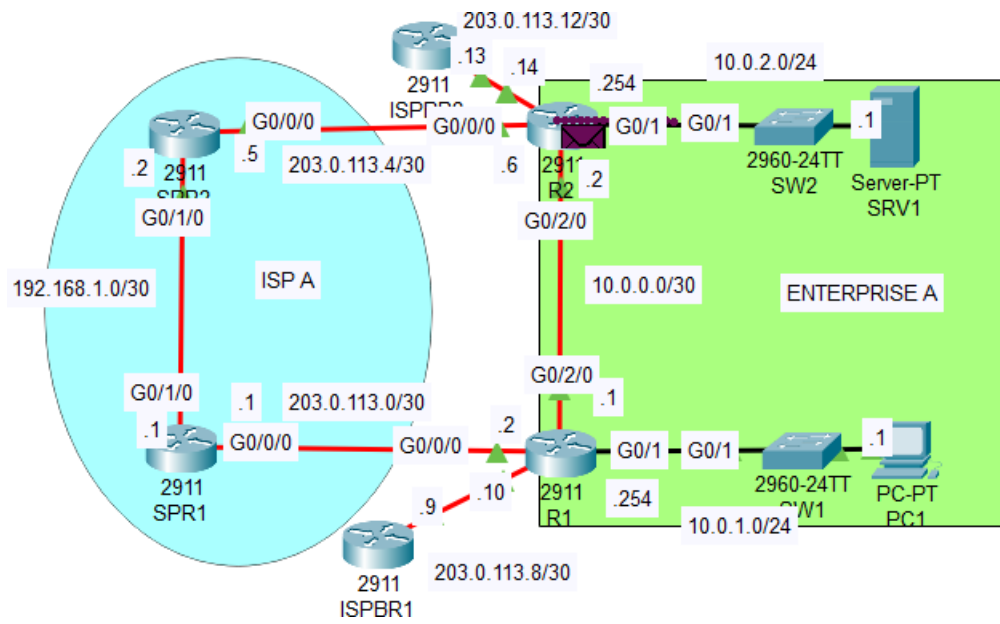
If SRV1 sends a reply back to R1, R2 will use the OSPF route to 10.0.1.0/24 via 10.0.0.1.

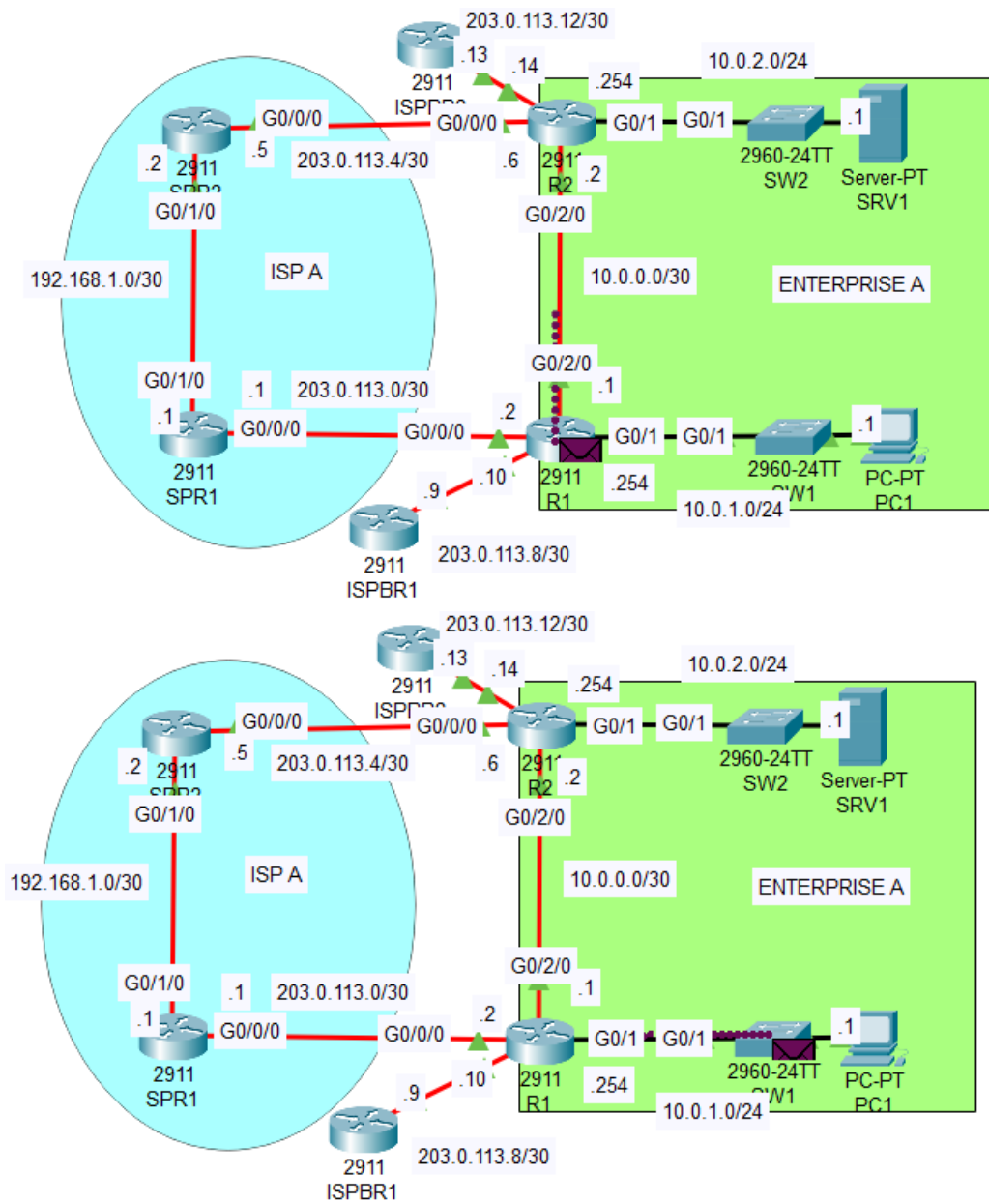




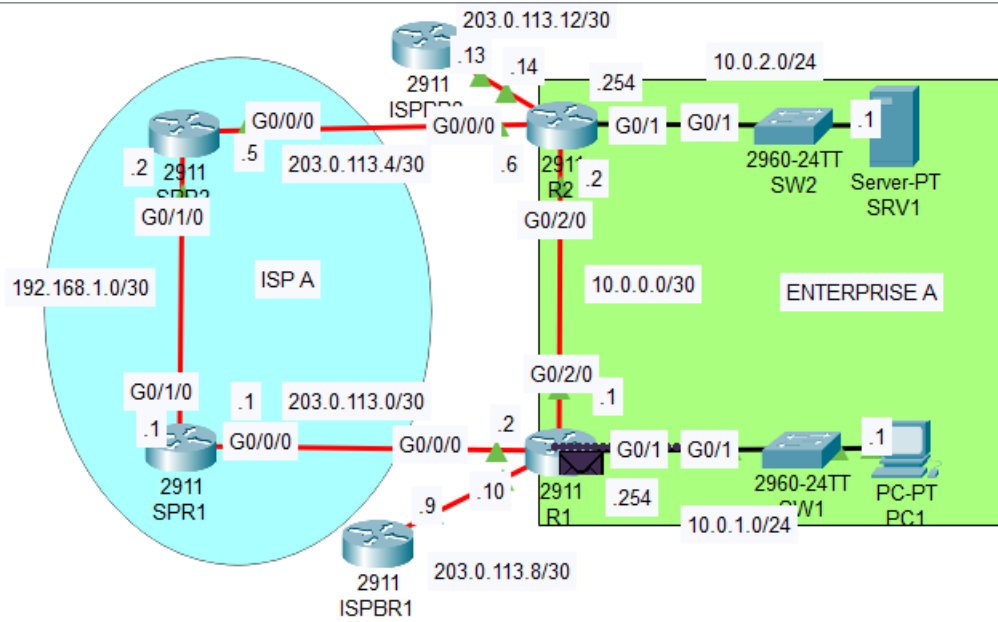
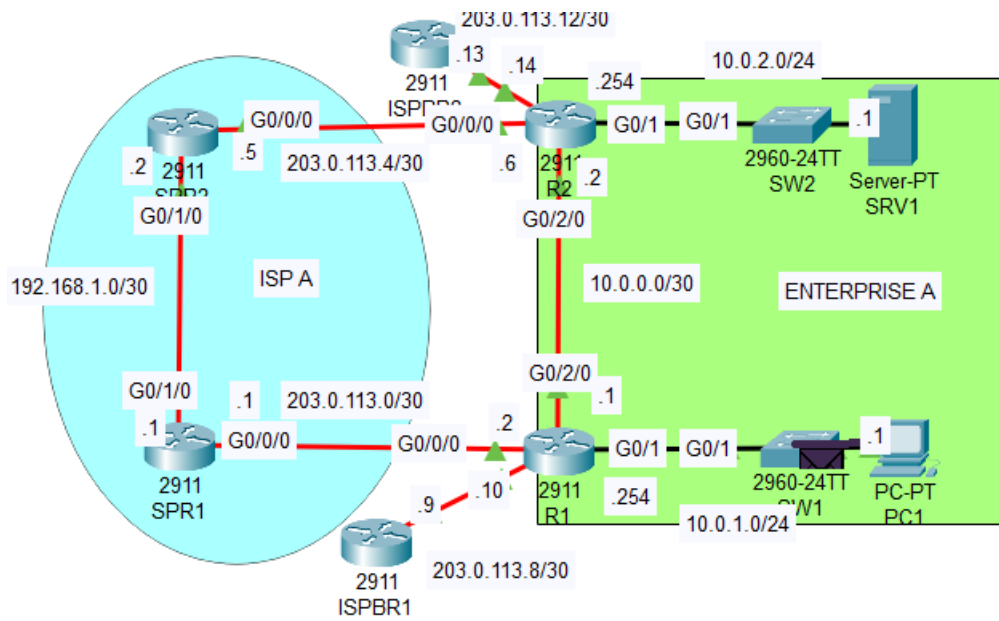


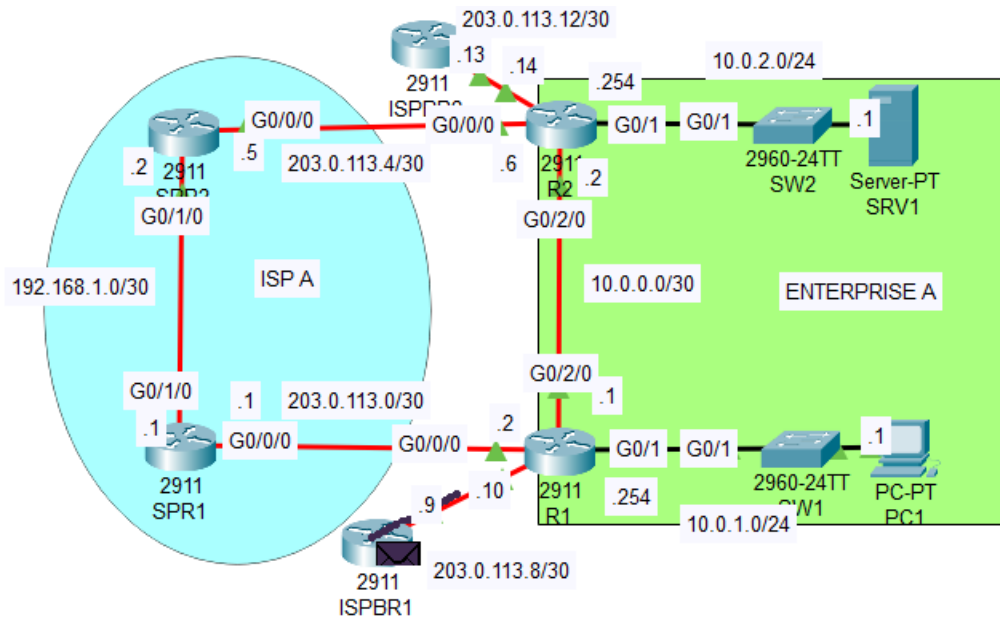






Ping from PC1 to SRV1





2. Configure floating static routes on R1 and R2 that allow PC1 to reach SRV1 if the link between R1 and R2 fails. Do the routes enter the routing tables of R1 and R2?

```
R1(config)#ip route 10.0.2.0 255.255.255.0 203.0.113.1 111
```

```
R1(config)#do sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is 203.0.113.9 to network 0.0.0.0
```

```
10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
C    10.0.0.0/30 is directly connected, GigabitEthernet0/2/0
L    10.0.0.1/32 is directly connected, GigabitEthernet0/2/0
C    10.0.1.0/24 is directly connected, GigabitEthernet0/1
L    10.0.1.254/32 is directly connected, GigabitEthernet0/1
O    10.0.2.0/24 [110/2] via 10.0.0.2, 00:40:00, GigabitEthernet0/2/0
203.0.113.0/24 is variably subnetted, 4 subnets, 2 masks
C    203.0.113.0/30 is directly connected, GigabitEthernet0/0/0
L    203.0.113.2/32 is directly connected, GigabitEthernet0/0/0
C    203.0.113.8/30 is directly connected, GigabitEthernet0/1/0
L    203.0.113.10/32 is directly connected, GigabitEthernet0/1/0
S*   0.0.0.0/0 [1/0] via 203.0.113.9
```

Since the learned OSPF route has an administrative distance of 110, the static route to be configured needs to be set with a higher administrative distance, like 111, to be placed in a floating state. The floating static route will not appear in the show ip route command.

```

R2#
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip route 10.0.1.0 255.255.255.0 203.0.113.5 111
R2(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

```

Gateway of last resort is 203.0.113.13 to network 0.0.0.0

```

      10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/2/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/2/0
O       10.0.1.0/24 [110/2] via 10.0.0.1, 00:40:00, GigabitEthernet0/2/0
C       10.0.2.0/24 is directly connected, GigabitEthernet0/1
L       10.0.2.254/32 is directly connected, GigabitEthernet0/1
      203.0.113.0/24 is variably subnetted, 4 subnets, 2 masks
C       203.0.113.4/30 is directly connected, GigabitEthernet0/0/0
L       203.0.113.6/32 is directly connected, GigabitEthernet0/0/0
C       203.0.113.12/30 is directly connected, GigabitEthernet0/1/0
L       203.0.113.14/32 is directly connected, GigabitEthernet0/1/0
S*      0.0.0.0/0 [1/0] via 203.0.113.13

```

3. Shut down the G0/2/0 interface of R1 or R2. Do the floating static routes enter the routing tables of R1 and R2? Ping from PC1 to SRV1 to confirm.

```

R2(config-if)#ip route 10.0.1.0 255.255.255.0 203.0.113.5 111
R2(config-if)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

```

Gateway of last resort is 203.0.113.13 to network 0.0.0.0

```

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
S       10.0.1.0/24 [111/0] via 203.0.113.5
C       10.0.2.0/24 is directly connected, GigabitEthernet0/1
L       10.0.2.254/32 is directly connected, GigabitEthernet0/1
      203.0.113.0/24 is variably subnetted, 4 subnets, 2 masks
C       203.0.113.4/30 is directly connected, GigabitEthernet0/0/0
L       203.0.113.6/32 is directly connected, GigabitEthernet0/0/0
C       203.0.113.12/30 is directly connected, GigabitEthernet0/1/0
L       203.0.113.14/32 is directly connected, GigabitEthernet0/1/0
S*      0.0.0.0/0 [1/0] via 203.0.113.13

```

```
R1(config)#do sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
* - candidate default, U - per-user static route, o - ODR  
P - periodic downloaded static route
```

```
Gateway of last resort is 203.0.113.9 to network 0.0.0.0
```

```
10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks  
C    10.0.1.0/24 is directly connected, GigabitEthernet0/1  
L    10.0.1.254/32 is directly connected, GigabitEthernet0/1  
S    10.0.2.0/24 [111/0] via 203.0.113.1  
203.0.113.0/24 is variably subnetted, 4 subnets, 2 masks  
C    203.0.113.0/30 is directly connected, GigabitEthernet0/0/0  
L    203.0.113.2/32 is directly connected, GigabitEthernet0/0/0  
C    203.0.113.8/30 is directly connected, GigabitEthernet0/1/0  
L    203.0.113.10/32 is directly connected, GigabitEthernet0/1/0  
S*   0.0.0.0/0 [1/0] via 203.0.113.9
```

Yes